Frequency Dependence of Skin Depth

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Abstract

The Skin Depth Experiment examines the frequency dependence of magnetic-field penetration (skin depth, δ) in metallic cylinders by measuring the induced RMS voltage in a secondary coil as a function of the driving frequency. Following the method of Wiederick and Gauthier (1983), the ratio V/ν was plotted against $\nu^{-1/2}$ for four metals, producing linear relationships whose slope and intercept represent the internal and external magnetic flux components, respectively. The results confirm the expected trend that the slope increases with resistivity, while the strong deviation observed for steel reflects the influence of ferromagnetism and its frequency-dependent permeability. The analysis supports the theoretical dependence $\delta \propto \nu^{-1/2}$ for non-magnetic conductors and provides a quantitative comparison with literature conductivities, under the assumption that skin depth varies inversely with the square root of electrical conductivity. Using copper as a reference, deviations of 71% and 81% were obtained for aluminium and brass, respectively, with steel exhibiting a substantially larger discrepancy.

1 Introduction

In a conductor subjected to an alternating magnetic field, induced eddy currents act to oppose the changing flux, following Lenz's law. This interaction causes the internal magnetic field to diminish exponentially with depth, a process known as the *skin effect*. The characteristic penetration distance, where the field and current density decrease to $1/e$ of their surface magnitudes, is defined as the *skin depth* (δ), expressed as:

$$\delta = \sqrt{\frac{2\rho}{\mu\omega}}, \tag{1}$$

where ρ is the resistivity, μ the magnetic permeability, and $\omega = 2\pi\nu$ the angular frequency of the applied field (Griffiths 2017). As the frequency increases, the field penetration decreases, leading to an apparent rise in effective resistance.

2 Theory

When a cylindrical conductor is subjected to an alternating magnetic field generated by a solenoid, the total magnetic flux linking a secondary pick-up coil is given by:

$$\Phi_{\text{tot}} = \Phi_{\text{ext}} + \Phi_{\text{int}}, \tag{2}$$

where Φ_{ext} represents the flux external to the sample, and Φ_{int} the flux penetrating the conductor. The induced voltage is therefore:

$$V = N\frac{d\Phi_{\text{tot}}}{dt} = N\frac{d(\Phi_{\text{ext}} + \Phi_{\text{int}})}{dt}. \tag{3}$$

Under conditions of linear response, Wiederick and Gauthier (1983) demonstrated that plotting V/ν against $\nu^{-1/2}$ yields a linear relationship:

$$\frac{V}{\nu} = A + B\nu^{-1/2}, \tag{4}$$

where the intercept $A = k\Phi_{\text{ext}}$ represents the external flux contribution, and the slope $B = kC\sqrt{\rho/\mu}$ corresponds to the flux penetrating the conductor. Since resistivity and conductivity are inversely related,

$$\rho = \frac{1}{\sigma}, \tag{5}$$

it follows that

$$B \propto \sqrt{\frac{1}{\mu\sigma}}.$$

For non-magnetic materials where $\mu \approx \mu_0$, this reduces to $B \propto 1/\sqrt{\sigma}$, demonstrating that materials with higher conductivity exhibit smaller slopes. The constants k and C depend on the coil geometry and field uniformity (Reitz, Milford, and Christy 1993). This expression provides a direct means of testing the theoretical scaling $\delta \propto \nu^{-1/2}$ and comparing materials with differing ρ and μ.

3 Methodology

3.1 Apparatus

The experimental arrangement followed the setup described in Wiederick & Gauthier (1983) this consisted of a long solenoid (primary coil) connected in series with a low-distortion function generator and a precision resistor, and a secondary pick-up coil wound around the conducting cylinder. Cylindrical samples of copper, aluminium, brass, and steel were used, each closely fitting within the solenoid bore. The induced RMS voltage was measured using a high-impedance digital RMS voltmeter.

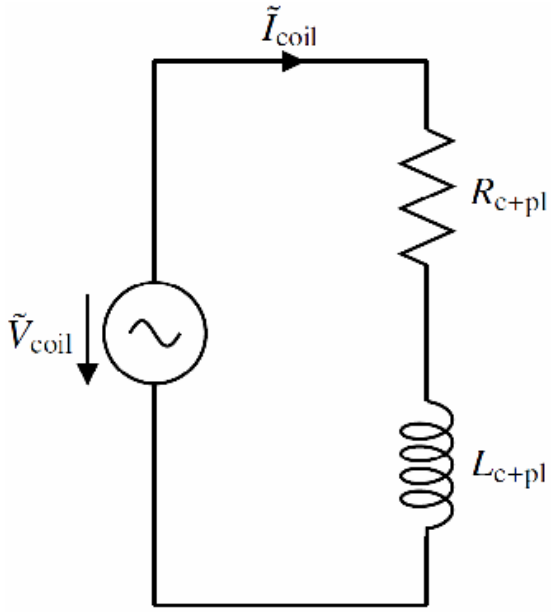


Figure 1: Equivalent circuit model of the primary excitation coil used in the skin-depth experiment. The alternating voltage source $\tilde{V}_{\text{coil}}$ drives a current $\tilde{I}_{\text{coil}}$ through the combined resistance $R_{\text{c+pl}}$ and inductance $L_{\text{c+pl}}$ of the coil and the connected wiring. This representation models the primary solenoid circuit used to generate the oscillating magnetic field that induces voltage in the pickup coil.

3.2 Procedure

Each metal cylinder was then inserted in turn, ensuring consistent positioning within the solenoid with equal lengths hanging over either side and that the crocodile clips were attached to a stripped area of the wire as the wires are coated in enamel. Weighted least-squares regression was applied to graphs modelled on Equation 4 to determine A and B with uncertainties. All data analysis is detailed as Python Code in the Appendix.

The driving current through the primary coil was maintained at $19 \pm 0.05\,\text{mA}$ using the series resistor, while the driving frequency was varied from 7 kHz to 61 kHz in increments of 3 kHz. Cylinder dimensions were recorded using a ruler with a precision of 0.05 cm.

3.3 Error Analysis

Instrument specifications were: frequency $\pm(1\% + 1\,\text{Hz})$, current $\pm 0.05\,\text{mA}$, and voltage $\pm 0.005\,\text{mV}$. Uncertainties in A and B were taken from the fit covariance matrix (weighted least squares with $\sigma_{V/\nu}$ as weights; see Appendix). Systematic effects were small compared to fit uncertainties and were not added in quadrature. Relative uncertainties in B were $<5\%$ across samples. Two runs were discarded due to enamel contact errors. At low frequency ($< 10\,\text{kHz}$) increased scatter is consistent with lower SNR and coil background inductance. High-frequency outliers that broke linearity were masked unmasked plots may be found in the Appendix.

Slightly larger scatter was observed at the lowest frequencies points (7,10,13 kHz), most likely this is due to the reduced signal-to-noise ratio and increased contribution from the background inductance of the coils. However, these points were determined not to cause Considerable error to the line of best fit for each plot. Points at high frequency however broke linearity severely so were masked, See unmasked graphs may be found bottom of appendix.

Uncertainty Propagation

To quantify the uncertainty in the fitted parameters A and B from the linear model descirbed in Equation 4 we propagate the experimental measurement uncertainties in V and ν using standard Gaussian error propagation (see Hughes & Hase, Sec. 2.8).

In general $y = f(x_1, x_2, \ldots, x_n)$ with independent uncertainties σ_{x_i}, the uncertainty in y is

$$\sigma_y^2 = \sum_i \left(\frac{\partial f}{\partial x_i}\right)^2 \sigma_{x_i}^2.$$

In this Experiment

$$f(V, \nu) = \frac{V}{\nu},$$

so the propagated uncertainty on each data point is

$$\sigma_{V/\nu}^2 = \left(\frac{1}{\nu}\right)^2 \sigma_V^2 + \left(\frac{V}{\nu^2}\right)^2 \sigma_\nu^2.$$

The uncertainty $\sigma_{V/\nu}$ is used as the weighting parameter see apendix.

From the least squares fit uncertainty can be dervied on A and B below as,

$$\sigma_A = \sqrt{C_{AA}}, \qquad \sigma_B = \sqrt{C_{BB}}, \qquad C_{AB} = \mathrm{cov}(A, B).$$

To propagate these to the skin depth δ or conductivity σ, we use

$$\delta = kB^2 \quad \Rightarrow \quad \sigma_\delta = \left|\frac{\partial \delta}{\partial B}\right| \sigma_B = 2kB\sigma_B,$$

where k is a proportionality constant see Theory.

The final reported values are quoted as

$$A = (A_{\text{fit}} \pm \sigma_A), \qquad B = (B_{\text{fit}} \pm \sigma_B),$$

and any derived parameters include the propagated uncertainties from the sigma A and B accordingly.

4 Results

The fitted intercept (A) and slope (B) values obtained from linear regression of V/ν versus $\nu^{-1/2}$ are visually shown in Figures 2-6 and summarised in Table 1.

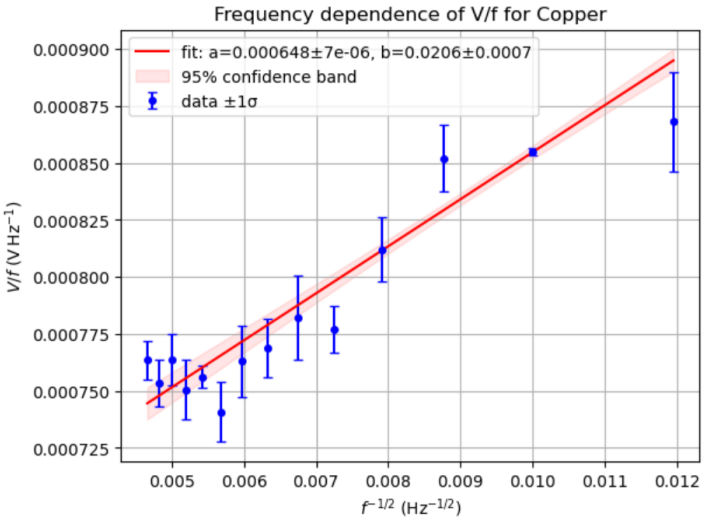


Figure 2: Plot of V/ν against $\nu^{-1/2}$ for the copper cylinder, showing the linear fit used to extract parameters A and B.

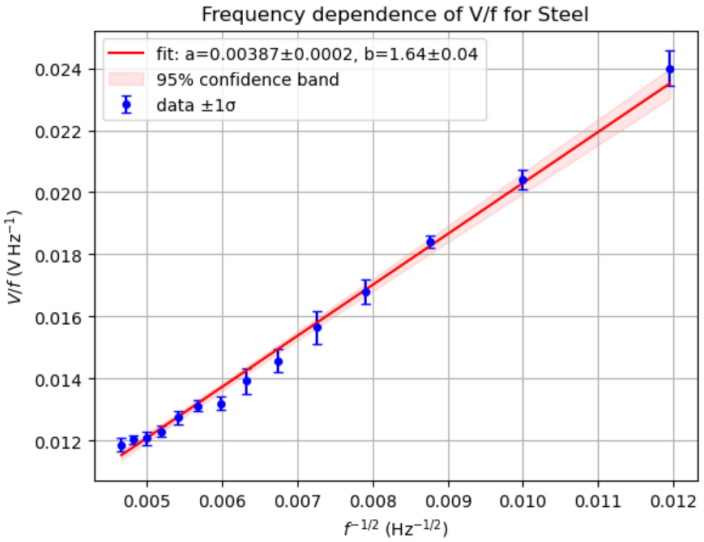


Figure 3: Plot of V/ν against $\nu^{-1/2}$ for the steel cylinder.

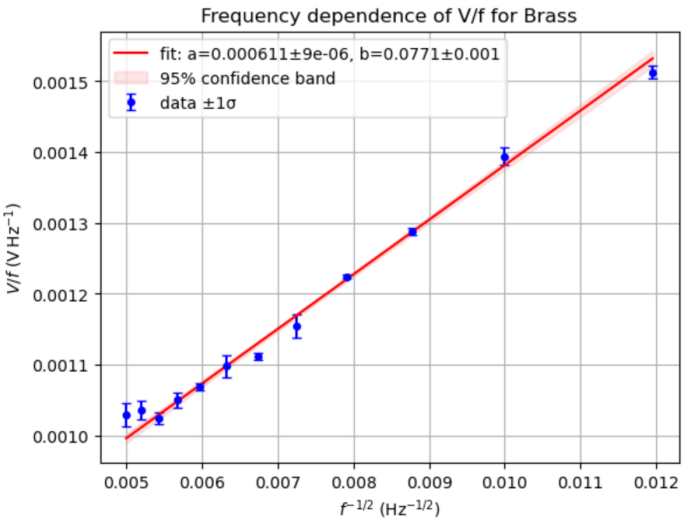


Figure 4: Plot of V/ν against $\nu^{-1/2}$ for the brass cylinder.

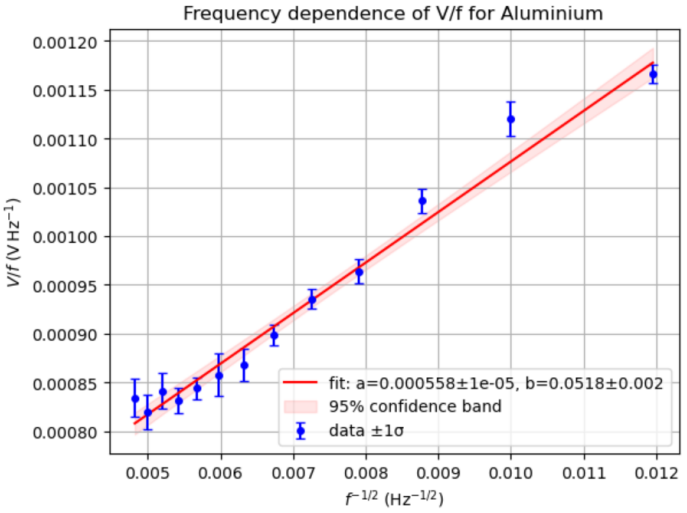


Figure 5: Plot of V/ν against $\nu^{-1/2}$ for the aluminium cylinder.

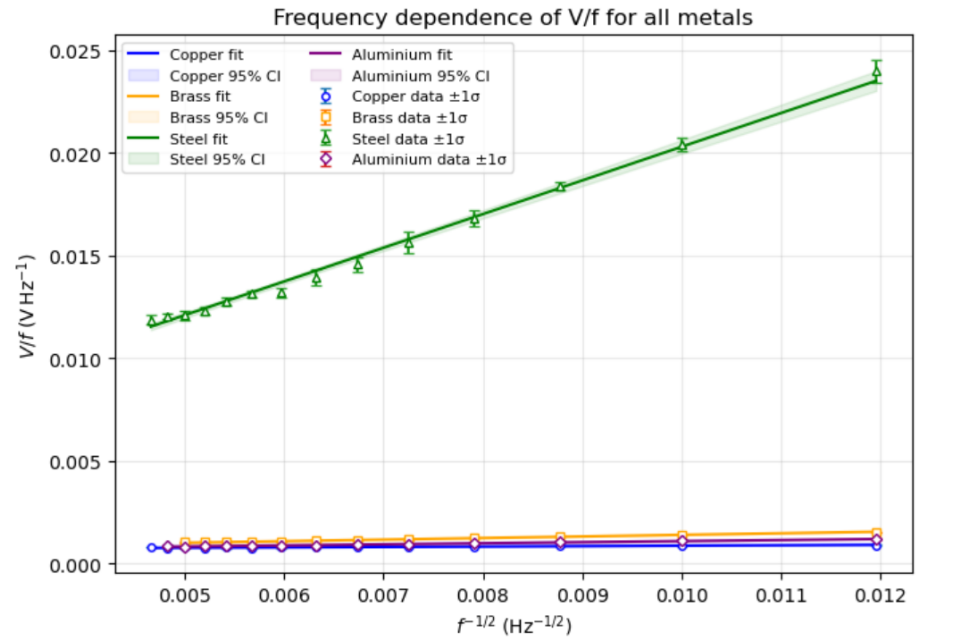


Figure 6: Combined comparison of V/ν versus $\nu^{-1/2}$ for all metals. The differing slopes illustrate the relationship between conductivity, magnetic properties, and skin-depth behaviour.

Table 1: Fitted parameters for $V/\nu = A + B\nu^{-1/2}$ for different metals.

Material	A / V Hz^{-1}×10^{-5}	B / V Hz$^{-1.5}$
Brass	$(6.11 \pm 0.90) \times 10^{-5}$	0.0771 ± 0.0010
Copper	$(6.48 \pm 0.70) \times 10^{-5}$	0.0206 ± 0.0007
Aluminium	$(5.58 \pm 1.00) \times 10^{-5}$	0.0518 ± 0.0020
Steel	$(3.87 \pm 0.20) \times 10^{-3}$	1.64 ± 0.04

These values were taken from figures 2-5 below. The intercept A corresponds to the external magnetic flux linking the secondary coil, while B represents the component associated with field penetration into the conductor. The B values normalised to copper were:

$$\frac{B_{\text{brass}}}{B_{\text{Cu}}} = 3.7, \qquad \frac{B_{\text{Al}}}{B_{\text{Cu}}} = 2.5, \qquad \frac{B_{\text{steel}}}{B_{\text{Cu}}} = 79.6.$$

Room-temperature conductivities taken from literature are:

$$\sigma_{\text{Cu}} = 5.6 \times 10^7\,\text{S m}^{-1}, \quad \sigma_{\text{Al}} = 2.6 \times 10^7\,\text{S m}^{-1}, \quad \sigma_{\text{brass}} = 1.3 \times 10^7\,\text{S m}^{-1}$$

(Wiederick and Gauthier 1983). From Equation 1 these values of skin depth which are proportional to B which is tabulated above are also proportional to the inverse square root of the conductivity values above when normalised to copper.Table 2 presents the measured and theoretical ratios of B/B_{Cu} alongside the percentage deviation. The theoretical ratios were obtained from $\sqrt{\sigma_{\text{Cu}}/\sigma}$ using standard room-temperature conductivities (**crc2024**).

Table 2: Comparison of measured B ratios with theoretical expectations based on literature conductivities.

Material	Measured B/B_{Cu}	Theoretical $\sqrt{\sigma_{\mathrm{Cu}}/\sigma}$	Deviation / %
Copper	1.00	1.00	0
Aluminium	2.52	1.47	71
Brass	3.74	2.07	81
Steel	79.6	$\gg 10$	–

5 Discussion

The results show a clear linear relationship between V/ν and $\nu^{-1/2}$ for all samples, in strong agreement with the theoretical prediction of the skin-depth model. This confirms that the experiment successfully captured the expected frequency dependence of magnetic-field penetration in good conductors. The linearity across different materials demonstrates that the method is reliable for comparing metals of varying conductivity and magnetic properties within the studied frequency range.

For copper, the smallest slope B corresponds to the greatest skin depth, consistent with its high conductivity and minimal energy loss. Aluminium and brass produced progressively larger B values, reflecting their higher resistivities and shallower field penetration. The observed ordering,

$$B_{\mathrm{Cu}} < B_{\mathrm{Al}} < B_{\mathrm{brass}},$$

matches the ranking of conductivities reported in the literature (Wiederick and Gauthier 1983; Tipler and Mosca 2007). Although the absolute values deviate from theoretical expectations by 71 % (aluminium) and 81 % (brass), as shown in Table 2, the relative trend agrees with $B \propto 1/\sqrt{\sigma}$. These discrepancies likely arise from geometric uncertainties, non-uniform field coupling, and finite-length effects in the solenoid—factors not accounted for in the idealised analytical model.

Steel, in contrast, shows a dramatic departure from the non-magnetic behaviour. Both the slope B and intercept A are orders of magnitude larger than for copper, aluminium, or brass. This is attributed to the strong frequency dependence of the magnetic permeability of steel (Jackson 1999; B. I. Bleaney and B. Bleaney 1976). At 7–61 kHz, the permeability becomes complex, $\mu = \mu' - i\mu''$, where μ'' represents energy loss from domain-wall motion. These magnetic losses, combined with steel's relatively high conductivity, cause strong field exclusion and deviate in behaviour from the simple proportionality $B \propto \sqrt{\rho/\mu}$ (Reitz, Milford, and Christy 1993). The large intercept also suggests enhanced mutual inductance between the coils due to the ferromagnetic core. Despite this deviation, the data still exhibits linearity with $\nu^{-1/2}$, showing that the skin-depth model qualitatively holds even when the assumptions of constant μ break down and actually in the unmasked

code steel holds the best to linearity.

Overall, the experiment confirms the predicted inverse-square-root dependence of skin depth on frequency for non-magnetic metals and clearly illustrates the model's breakdown in ferromagnetic materials. The measured deviations from theory are well explained by practical imperfections and material-specific magnetic effects.

Future improvements could include testing another ferromagnetic core to confirm that magnetic permeability is the primary cause of steel's deviation, automating data acquisition to minimise reading error (although this was minor compared to overall data fluctuations), and monitoring the driving current with an oscilloscope to verify amplitude constancy. Extending measurements to lower frequencies (1–5 kHz) would help capture the transition where steel behaves more linearly before magnetic losses dominate. Most importantly, repeating the experiment multiple times would reduce statistical uncertainty and tighten the confidence intervals; approximately ten independent runs per sample would be required to achieve a 95% confidence level under the assumption of random, Gaussian-distributed errors.

Overall, the experiment successfully demonstrates the inverse-square-root dependence of skin depth on frequency for good conductors with relatively strong concurrence with literature values of conductivity provided, while the steel data serve as a clear indication of the model's limits in ferromagnetic regimes.

6 Conclusion

The frequency dependence of skin depth was investigated through induced voltage measurements in conducting cylinders. The plots of V/ν versus $\nu^{-1/2}$ produced linear relationships for all materials, confirming the theoretical model of electromagnetic field diffusion. For Copper, Aluminium, and Brass, the slopes scaled consistently with resistivity, verifying the relation $\delta \propto \nu^{-1/2}$. Steel deviated strongly due to its complex permeability and magnetic losses, illustrating the breakdown of the simple skin-depth expression in ferromagnetic materials and highlighting the limitations of our simple assumption.

The experimental results align closely with established literature conductivities and highlight the sensitivity of skin-depth phenomena to both electrical and magnetic material properties. Within experimental uncertainty, the findings demonstrate that the induced-voltage method provides a reliable means of probing electromagnetic field penetration in conducting media.

A Appendix

The following appendix contains the exported Jupyter Notebook. This Python code both stores the raw data and is used for data analysis, curve fitting, and figure generation for this experiment.

Store Raw Data

In [209…
```
import numpy as np
import matplotlib.pyplot as plt
```

In [90]:
```
#Steel data
mvs1 = (164.16,200.69,237.51,262.51,283.54,311.57,338.12,366.32,397.52,427.1,445
mvs2 = (163.95,200.88,235.77,262.60,291.86,312.84,337.7,360.9,405.16,425.07,452.
mvs3 = (175.87,210.53,244.50,280.93,317.11,337.2,368.01,381.25,417.31,447.00,467
khz=(7,10,13,16,19,22,25,28,31,34,37,40,43,46,49,52,55,58,61)
#Aluminium data
mva1 = (8.15,11.41,13.29,15.38,18.11,19.95,22.27,24.32,26.05,28.21,31.58,33.10,3
mva2 =(8.28,11.34,13.78,15.78,17.76,20.07,21.94,24.89,26.81,29.07,31.98,33.82,37
mva3 = (8.06,10.85,13.34,15.10,17.46,19.3,20.9,22.86,25.63,27.55,29.82,31.42,34.
#Brass data
mvb1 = (10.65,13.73,16.63,19.58,21.42,24.56,26.69,29.67,32.16,34.46,37.50,40.01,
mvb2 = (10.46,13.96,16.73,19.54,21.92,24.26,27.71,29.95,33.22,35.38,39.14,42.26,
mvb3 = (10.65,14.14,16.87,19.65,22.47,24.54,27.94,30.15,32.27,34.69,38.37,41.33,
#Copper data
mvc1 = (5.89,8.54,10.87,12.56,14.37,16.58,18.57,20.62,22.26,25.38,26.86,29.89,31
mvc2 = (6.38,8.58,10.91,13.30,14.95,17.07,19.55,21.32,22.96,25.96,27.98,30.35,32
mvc3 = (5.96,8.53,11.45,13.12,14.96,17.97,19.53,22.15,23.67,25.79,28.47,31.42,32
```

Data Manipulation

Aluminium masked results

In [200…
```
# Data preparation
mva3 = np.array(mva3)
mva2 = np.array(mva2)
mva1 = np.array(mva1)

#Derive Mean, Standard deviation and error based on Gaussian Distribution
mvatot = np.vstack([mva1, mva2, mva3])
mvam = np.mean(mvatot, axis=0)
mvas = np.std(mvatot, axis=0, ddof=1)
mvae = mvas / np.sqrt(3)

#define variables to graph
xa = Hz**(-0.5)
ya = mvam / Hz
yea = mvae / Hz

# Fit model
def model(x, a, b):
    return a + b*x
#Masking points that break linear trend
mask_a = xa > 0.0048
xa, ya, yea = xa[mask_a], ya[mask_a], yea[mask_a]

#fitting weighted least squares line
popt, pcov = curve_fit(model, xa, ya, sigma=yea, absolute_sigma=True)
aa, ba = popt
```

```
saa, sba = np.sqrt(np.diag(pcov))
#define line specific to each metal
xa_line = np.linspace(xa.min(), xa.max(), 200)
ya_line = model(xa_line, aa, ba)

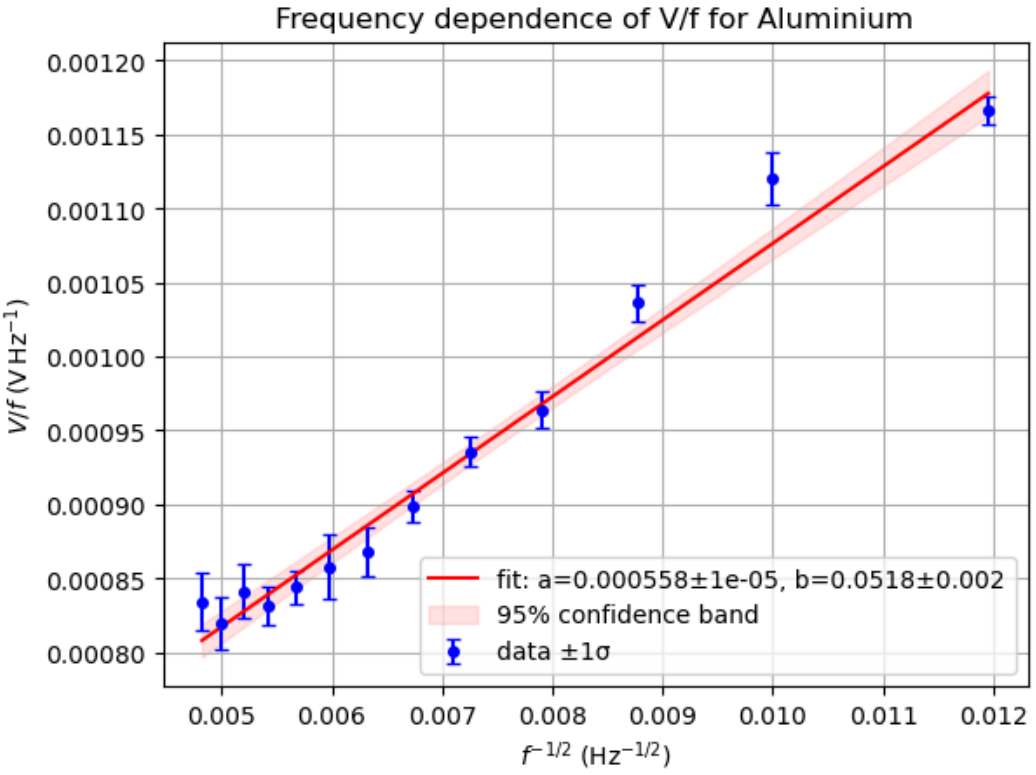
# Compute ±1 sigma fit uncertainty band
saa2, sba2 = pcov[0, 0], pcov[1, 1]
saab = pcov[0, 1]
err_a = np.sqrt(saa2 + (xa_line**2)*sba2 + 2*xa_line*saab)

# Compute 95% confidence band by multiplying standard error by 1.96
CIa = 1.96 * err_a

plt.errorbar(xa, ya, yerr=yea, fmt='o', capsize=3, ms=4,
             color='blue', label='data ±1σ')
plt.plot(xa_line, ya_line, 'r-',
         label=f'fit: a={aa:.3g}±{saa:.1g}, b={ba:.3g}±{sba:.1g}')

#Plot 95 percent confidence interval
plt.fill_between(xa_line, ya_line - CIa, ya_line + CIa,
                 color='red', alpha=0.1, label='95% confidence band')

#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for Aluminium')
plt.legend()
plt.grid(True)
plt.show()
```



Brass masked results

In [196…

```
# Data preparation
mvb3 = np.array(mvb3)
mvb2 = np.array(mvb2)
mvb1 = np.array(mvb1)

#Derive Mean, Standard deviation and error based on Gaussian Distribution
mvbtot = np.vstack([mvb1, mvb2, mvb3])
mvbm = np.mean(mvbtot, axis=0)
mvbs = np.std(mvbtot, axis=0, ddof=1)
mvbe = mvbs / np.sqrt(3)

#define variables to graph
xb = Hz**(-0.5)
yb = mvbm / Hz
yeb = mvbe / Hz

# Fit model
def model(x, a, b):
    return a + b*x
#Masking points that break linear trend
mask_b = xb > 0.0049
xb, yb, yeb = xb[mask_b], yb[mask_b], yeb[mask_b]

#fitting weighted least squares line
popt, pcov = curve_fit(model, xb, yb, sigma=yeb, absolute_sigma=True)
ab, bb = popt
sab, sbb = np.sqrt(np.diag(pcov))
#define line specific to each metal
xb_line = np.linspace(xb.min(), xb.max(), 200)
yb_line = model(xb_line, ab, bb)

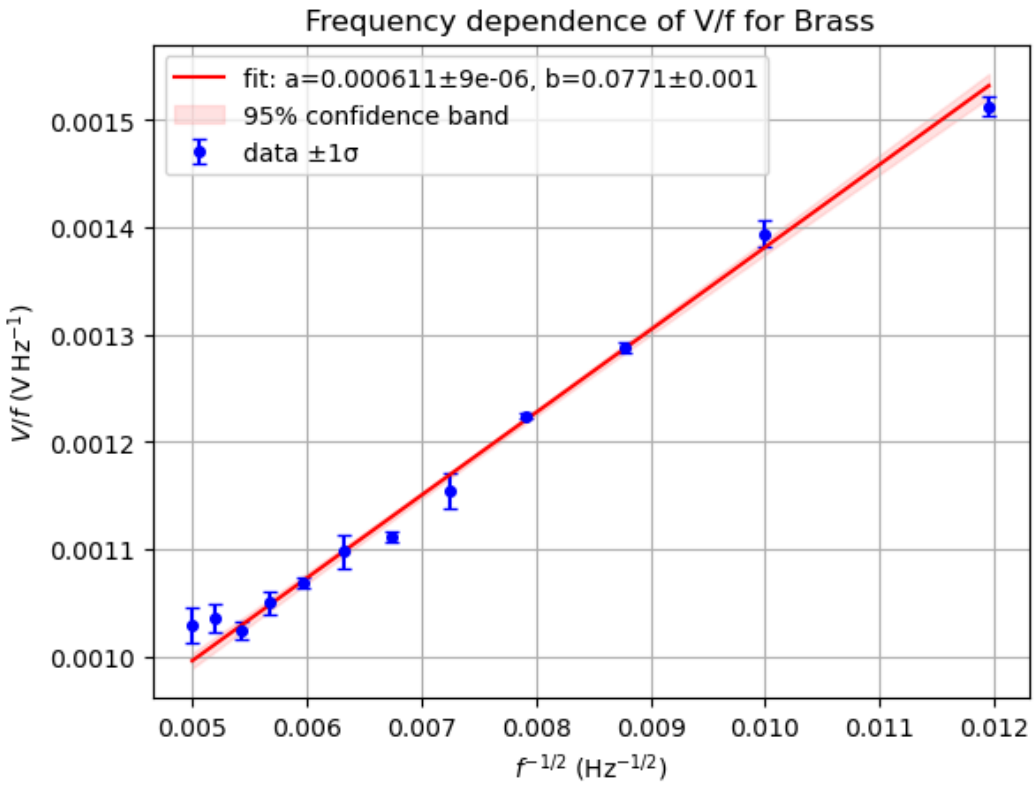
# Compute ±1 sigma fit uncertainty band
sab2, sbb2 = pcov[0, 0], pcov[1, 1]
sabb = pcov[0, 1]
err_b = np.sqrt(sab2 + (xb_line**2)*sbb2 + 2*xb_line*sabb)

# Compute 95% confidence band by multiplying standard error by 1.96
CIb = 1.96 * err_b

plt.errorbar(xb, yb, yerr=yeb, fmt='o', capsize=3, ms=4,
             color='blue', label='data ±1σ')
plt.plot(xb_line, yb_line, 'r-',
         label=f'fit: a={ab:.3g}±{sab:.1g}, b={bb:.3g}±{sbb:.1g}')

#Plot 95 percent confidence interval
plt.fill_between(xb_line, yb_line - CIb, yb_line + CIb,
                 color='red', alpha=0.1, label='95% confidence band')

#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for Brass')
plt.legend()
plt.grid(True)
plt.show()
```




```
0.0050541578791497245 1.4826402022437388
```

Steel masked results

In [189…

```
# Data preparation
mvs3 = np.array(mvs3)
mvs2 = np.array(mvs2)
mvs1 = np.array(mvs1)

#Derive Mean, Standard deviation and error based on Gaussian Distribution
mvstot = np.vstack([mvs1, mvs2, mvs3])
mvsm = np.mean(mvstot, axis=0)
mvss = np.std(mvstot, axis=0, ddof=1)
mvse = mvss / np.sqrt(3)

#define variables to graph
xs = Hz**(-0.5)
ys = mvsm / Hz
yes = mvse / Hz

# Fit model
def model(x, a, b):
    return a + b*x
#Masking points that break linear trend
mask_s = xs > 0.0046
xs, ys, yes = xs[mask_s], ys[mask_s], yes[mask_s]

#fitting weighted least squares line
popt, pcov = curve_fit(model, xs, ys, sigma=yes, absolute_sigma=True)
as_, bs = popt
sas, sbs = np.sqrt(np.diag(pcov))
#define line specific to each metal
```

```
xs_line = np.linspace(xs.min(), xs.max(), 200)
ys_line = model(xs_line, as_, bs)

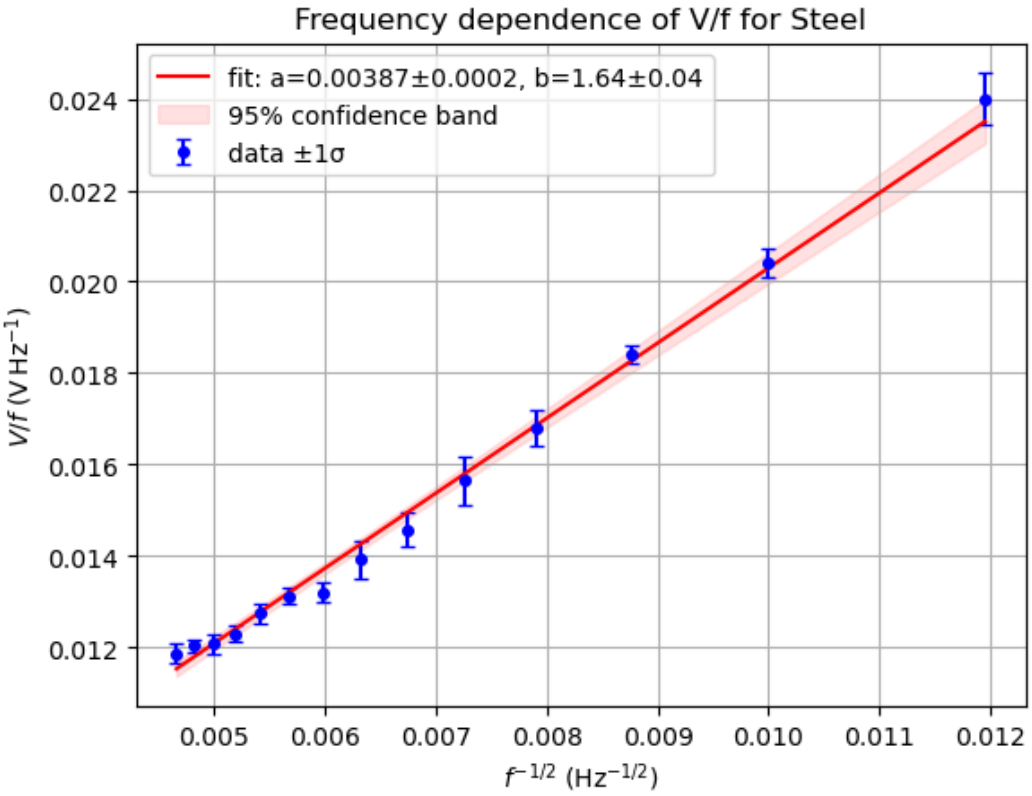
# Compute ±1 sigma fit uncertainty band
sas2, sbs2 = pcov[0, 0], pcov[1, 1]
sasb = pcov[0, 1]
err_s = np.sqrt(sas2 + (xs_line**2)*sbs2 + 2*xs_line*sasb)

# Compute 95% confidence band by multiplying standard error by 1.96
CIs = 1.96 * err_s

plt.errorbar(xs, ys, yerr=yes, fmt='o', capsize=3, ms=4,
             color='blue', label='data ±1σ')
plt.plot(xs_line, ys_line, 'r-',
         label=f'fit: a={as_:.3g}±{sas:.1g}, b={bs:.3g}±{sbs:.1g}')

#Plot 95 percent confidence interval
plt.fill_between(xs_line, ys_line - CIs, ys_line + CIs,
                 color='red', alpha=0.1, label='95% confidence band')

#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for Steel')
plt.legend()
plt.grid(True)
plt.show()
```



Copper Masked Results

In [197…

```
# Data preparation
mvc3 = np.array(mvc3)
mvc2 = np.array(mvc2)
mvc1 = np.array(mvc1)

#Derive Mean, Standard deviation and error based on Gaussian Distribution
mvctot = np.vstack([mvc1, mvc2, mvc3])
mvcm = np.mean(mvctot, axis=0)
mvcs = np.std(mvctot, axis=0, ddof=1)
mvce = mvcs / np.sqrt(3)

#define variables to graph
xc = Hz**(-0.5)
yc = mvcm / Hz
yec = mvce / Hz

# Fit model
def model(x, a, b):
    return a + b*x
#Masking points that break linear trend
mask_c = xc > 0.0046
xc, yc, yec = xc[mask_c], yc[mask_c], yec[mask_c]

#fitting weighted least squares line
popt, pcov = curve_fit(model, xc, yc, sigma=yec, absolute_sigma=True)
ac, bc = popt
sac, sbc = np.sqrt(np.diag(pcov))

#define line specific to each metal
xc_line = np.linspace(xc.min(), xc.max(), 200)
yc_line = model(xc_line, ac, bc)

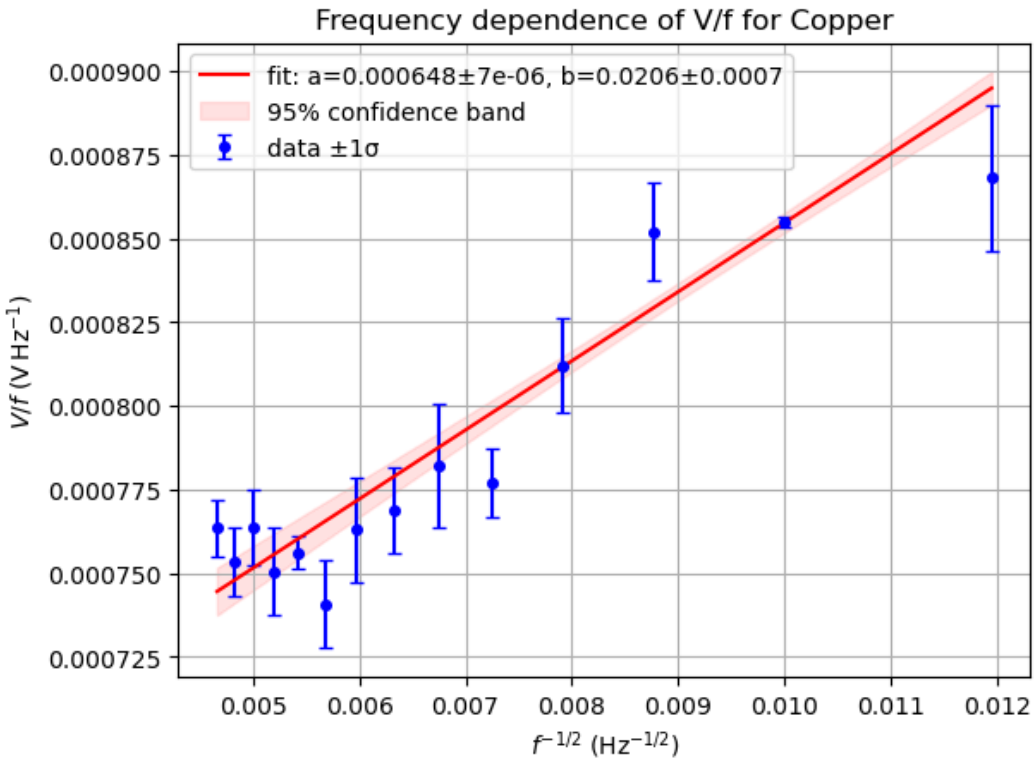
# Compute ±1 sigma fit uncertainty band
sac2, sbc2 = pcov[0, 0], pcov[1, 1]
sacb = pcov[0, 1]
err_c = np.sqrt(sac2 + (xc_line**2)*sbc2 + 2*xc_line*sacb)

# Compute 95% confidence band by multiplying standard error by 1.96
CIc = 1.96 * err_c

plt.errorbar(xc, yc, yerr=yec, fmt='o', capsize=3, ms=4,
             color='blue', label='data ±1σ')
plt.plot(xc_line, yc_line, 'r-',
         label=f'fit: a={ac:.3g}±{sac:.1g}, b={bc:.3g}±{sbc:.1g}')

#Plot 95 percent confidence interval
plt.fill_between(xc_line, yc_line - CIc, yc_line + CIc,
                 color='red', alpha=0.1, label='95% confidence band')

#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for Copper')
plt.legend()
plt.grid(True)
plt.show()
```



Combined masked results

In [195…

```
#Plot all metals together
plt.figure(figsize=(7,5))

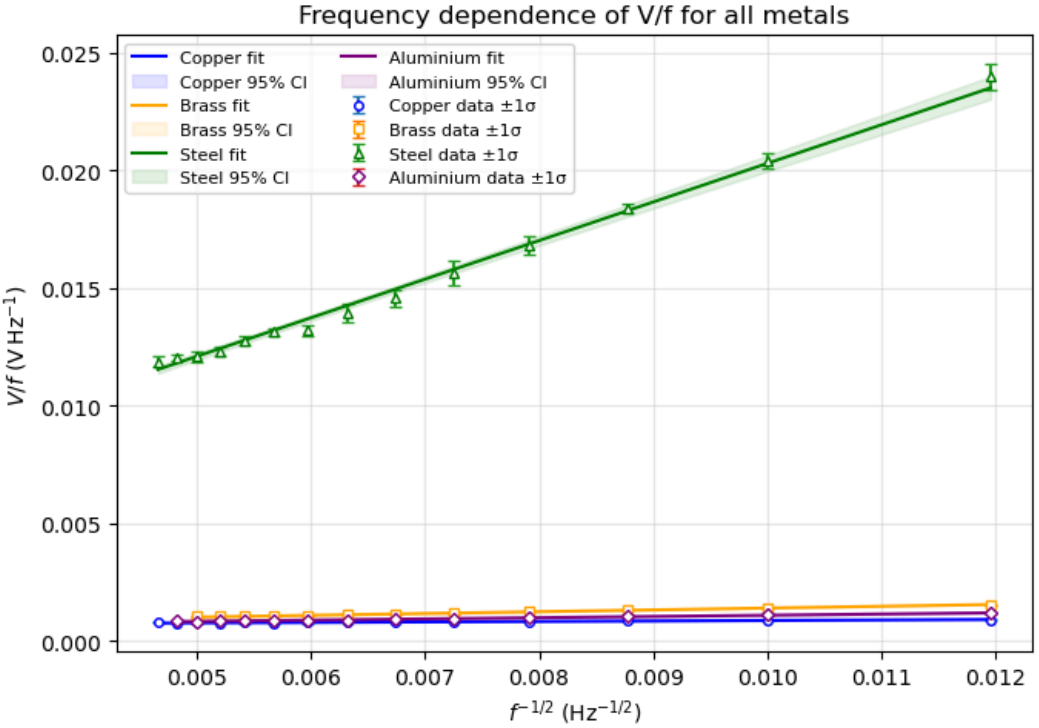
# Copper with error bars coloured and uncertaities
plt.errorbar(xc, yc, yerr=yec, fmt='o', capsize=3, ms=4,
             mfc='white', mec='blue', mew=1, label='Copper data ±1σ')
plt.plot(xc_line, yc_line, 'b-', label='Copper fit')
plt.fill_between(xc_line, yc_line - CIc, yc_line + CIc,
                 color='blue', alpha=0.1, label='Copper 95% CI')

# Brass with error bars coloured and uncertaities
plt.errorbar(xb, yb, yerr=yeb, fmt='s', capsize=3, ms=4,
             mfc='white', mec='orange', mew=1, label='Brass data ±1σ')
plt.plot(xb_line, yb_line, 'orange', label='Brass fit')
plt.fill_between(xb_line, yb_line - CIb, yb_line + CIb,
                 color='orange', alpha=0.1, label='Brass 95% CI')

# Steel with error bars coloured and uncertaities
plt.errorbar(xs, ys, yerr=yes, fmt='^', capsize=3, ms=4,
             mfc='white', mec='green', mew=1, label='Steel data ±1σ')
plt.plot(xs_line, ys_line, 'g-', label='Steel fit')
plt.fill_between(xs_line, ys_line - CIs, ys_line + CIs,
                 color='green', alpha=0.1, label='Steel 95% CI')

# Aluminium  with error bars coloured and uncertaities
plt.errorbar(xa, ya, yerr=yea, fmt='D', capsize=3, ms=4,
             mfc='white', mec='purple', mew=1, label='Aluminium data ±1σ')
plt.plot(xa_line, ya_line, 'purple', label='Aluminium fit')
plt.fill_between(xa_line, ya_line - CIa, ya_line + CIa,
                 color='purple', alpha=0.1, label='Aluminium 95% CI')
```

```
#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for all metals')
plt.legend(fontsize=8, ncol=2)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Raw graphs no Masking

In [201…

```
# Aluminium

# Data preparation
mva3 = np.array(mva3)
mva2 = np.array(mva2)
mva1 = np.array(mva1)

#Derive Mean, Standard deviation and error based on Gaussian Distribution
mvatot = np.vstack([mva1, mva2, mva3])
mvam = np.mean(mvatot, axis=0)
mvas = np.std(mvatot, axis=0, ddof=1)
mvae = mvas / np.sqrt(3)

#define variables to graph
xa = Hz**(-0.5)
ya = mvam / Hz
yea = mvae / Hz

# Fit model
def model(x, a, b):
    return a + b*x
```

```
#fitting weighted least squares line
popt, pcov = curve_fit(model, xa, ya, sigma=yea, absolute_sigma=True)
aa, ba = popt
saa, sba = np.sqrt(np.diag(pcov))

#define line specific to each metal
xa_line = np.linspace(xa.min(), xa.max(), 200)
ya_line = model(xa_line, aa, ba)

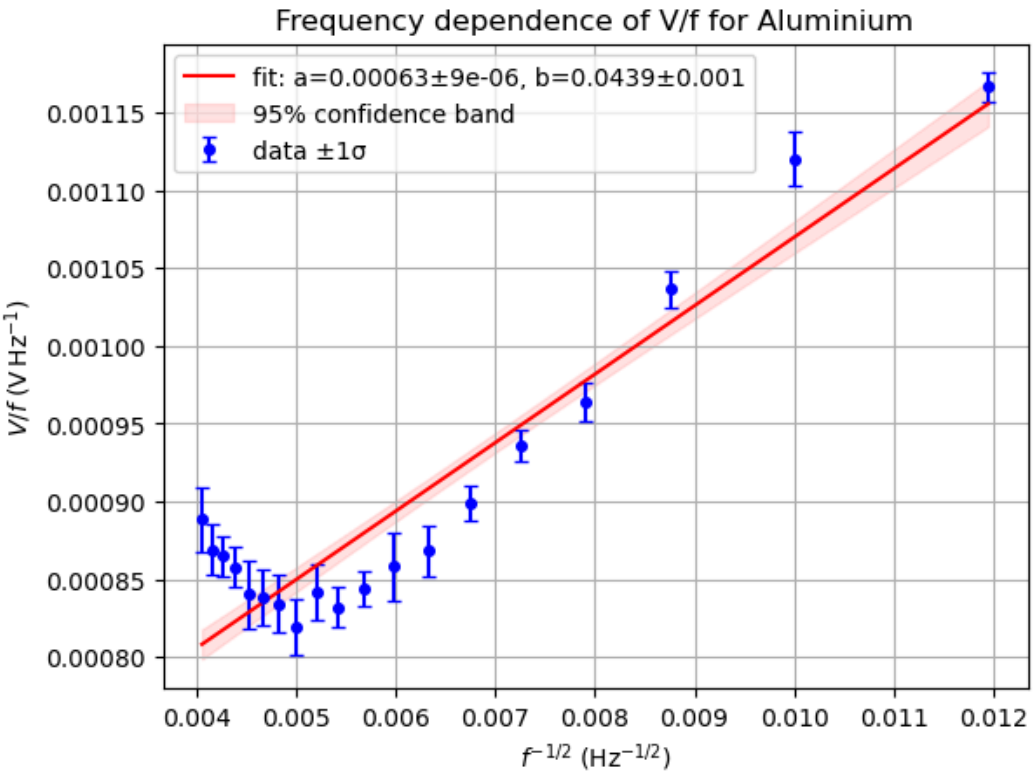
# Compute ±1 sigma fit uncertainty band
saa2, sba2 = pcov[0, 0], pcov[1, 1]
saab = pcov[0, 1]
err_a = np.sqrt(saa2 + (xa_line**2)*sba2 + 2*xa_line*saab)

# Compute 95% confidence band by multiplying standard error by 1.96
CIa = 1.96 * err_a

plt.errorbar(xa, ya, yerr=yea, fmt='o', capsize=3, ms=4,
             color='blue', label='data ±1σ')
plt.plot(xa_line, ya_line, 'r-',
         label=f'fit: a={aa:.3g}±{saa:.1g}, b={ba:.3g}±{sba:.1g}')

#Plot 95 percent confidence interval
plt.fill_between(xa_line, ya_line - CIa, ya_line + CIa,
                 color='red', alpha=0.1, label='95% confidence band')

#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for Aluminium')
plt.legend()
plt.grid(True)
plt.show()
```



In [202…

```
# Brass

# Data preparation
mvb3 = np.array(mvb3)
mvb2 = np.array(mvb2)
mvb1 = np.array(mvb1)

#Derive Mean, Standard deviation and error based on Gaussian Distribution
mvbtot = np.vstack([mvb1, mvb2, mvb3])
mvbm = np.mean(mvbtot, axis=0)
mvbs = np.std(mvbtot, axis=0, ddof=1)
mvbe = mvbs / np.sqrt(3)

#define variables to graph
xb = Hz**(-0.5)
yb = mvbm / Hz
yeb = mvbe / Hz

# Fit model
def model(x, a, b):
    return a + b*x

#fitting weighted least squares line
popt, pcov = curve_fit(model, xb, yb, sigma=yeb, absolute_sigma=True)
ab, bb = popt
sab, sbb = np.sqrt(np.diag(pcov))
#define line specific to each metal
xb_line = np.linspace(xb.min(), xb.max(), 200)
yb_line = model(xb_line, ab, bb)

# Compute ±1 sigma fit uncertainty band
sab2, sbb2 = pcov[0, 0], pcov[1, 1]
sabb = pcov[0, 1]
```

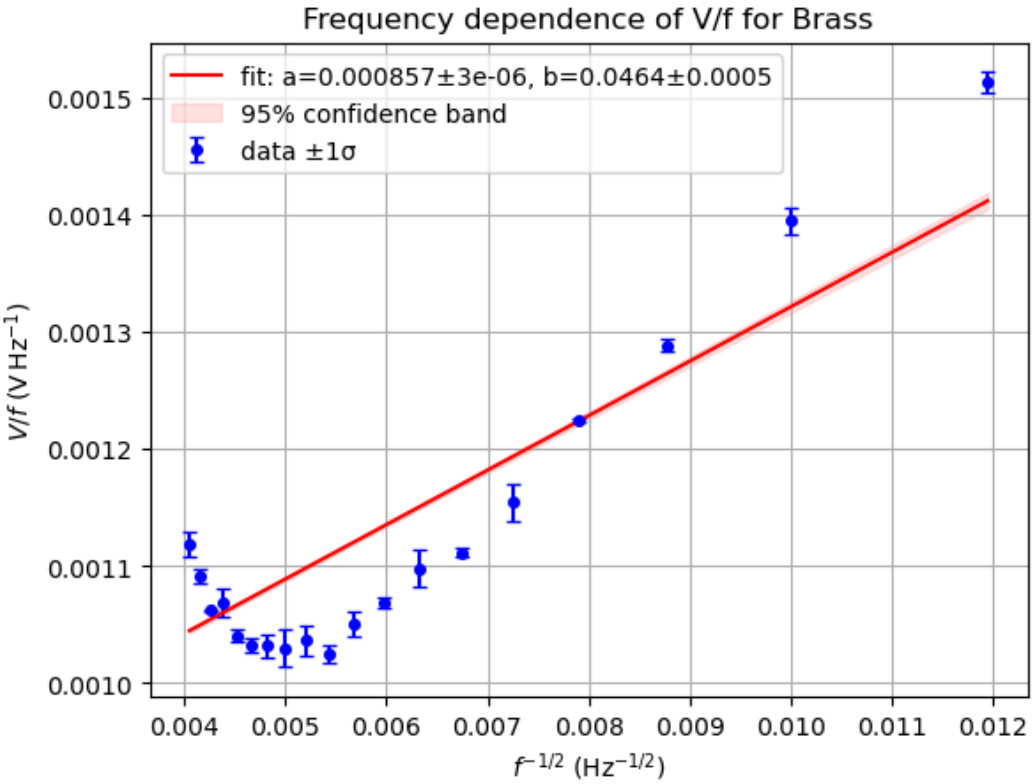
```
err_b = np.sqrt(sab2 + (xb_line**2)*sbb2 + 2*xb_line*sabb)

# Compute 95% confidence band by multiplying standard error by 1.96
CIb = 1.96 * err_b

plt.errorbar(xb, yb, yerr=yeb, fmt='o', capsize=3, ms=4,
             color='blue', label='data ±1σ')
plt.plot(xb_line, yb_line, 'r-',
         label=f'fit: a={ab:.3g}±{sab:.1g}, b={bb:.3g}±{sbb:.1g}')

#Plot 95 percent confidence interval
plt.fill_between(xb_line, yb_line - CIb, yb_line + CIb,
                 color='red', alpha=0.1, label='95% confidence band')

#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for Brass')
plt.legend()
plt.grid(True)
plt.show()
```



In [204…

```
#opper

# Data preparation
mvc3 = np.array(mvc3)
mvc2 = np.array(mvc2)
mvc1 = np.array(mvc1)

#Derive Mean, Standard deviation and error based on Gaussian Distribution
mvctot = np.vstack([mvc1, mvc2, mvc3])
mvcm = np.mean(mvctot, axis=0)
mvcs = np.std(mvctot, axis=0, ddof=1)
```

```
mvce = mvcs / np.sqrt(3)

#define variables to graph
xc = Hz**(-0.5)
yc = mvcm / Hz
yec = mvce / Hz

# Fit model
def model(x, a, b):
    return a + b*x

#fitting weighted least squares line
popt, pcov = curve_fit(model, xc, yc, sigma=yec, absolute_sigma=True)
ac, bc = popt
sac, sbc = np.sqrt(np.diag(pcov))

#define line specific to each metal
xc_line = np.linspace(xc.min(), xc.max(), 200)
yc_line = model(xc_line, ac, bc)

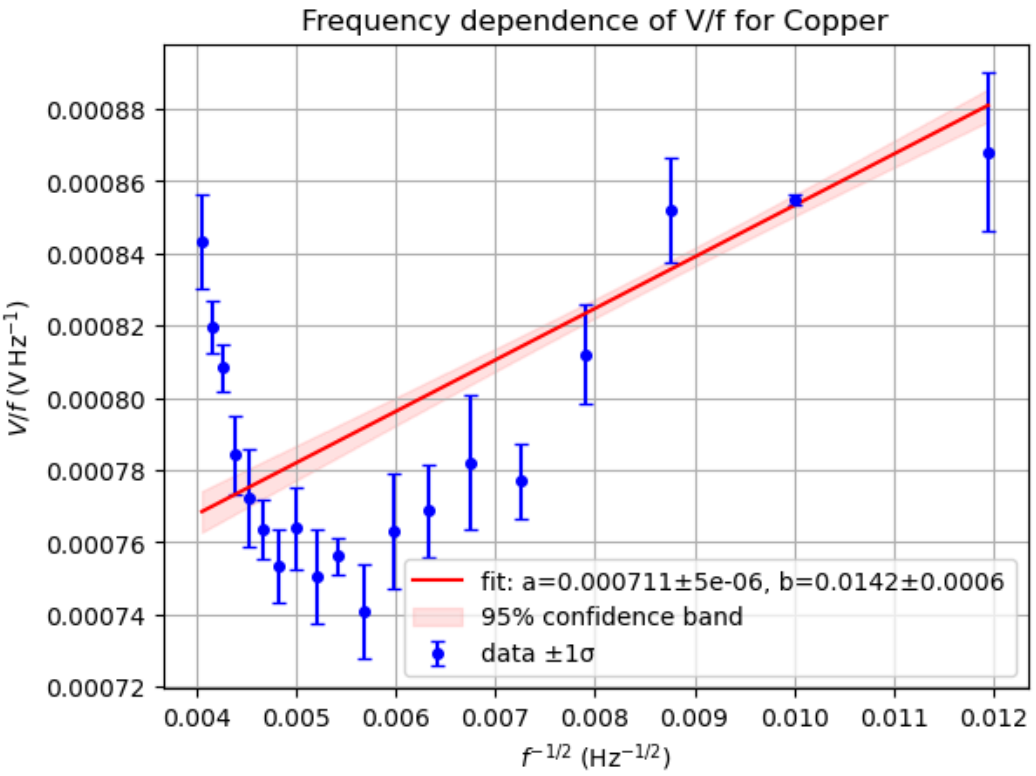
# Compute ±1 sigma fit uncertainty band
sac2, sbc2 = pcov[0, 0], pcov[1, 1]
sacb = pcov[0, 1]
err_c = np.sqrt(sac2 + (xc_line**2)*sbc2 + 2*xc_line*sacb)

# Compute 95% confidence band by multiplying standard error by 1.96
CIc = 1.96 * err_c

plt.errorbar(xc, yc, yerr=yec, fmt='o', capsize=3, ms=4,
             color='blue', label='data ±1σ')
plt.plot(xc_line, yc_line, 'r-',
         label=f'fit: a={ac:.3g}±{sac:.1g}, b={bc:.3g}±{sbc:.1g}')

#Plot 95 percent confidence interval
plt.fill_between(xc_line, yc_line - CIc, yc_line + CIc,
                 color='red', alpha=0.1, label='95% confidence band')

#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for Copper')
plt.legend()
plt.grid(True)
plt.show()
```



In [206…

```
# Data preparation
mvs3 = np.array(mvs3)
mvs2 = np.array(mvs2)
mvs1 = np.array(mvs1)

#Derive Mean, Standard deviation and error based on Gaussian Distribution
mvstot = np.vstack([mvs1, mvs2, mvs3])
mvsm = np.mean(mvstot, axis=0)
mvss = np.std(mvstot, axis=0, ddof=1)
mvse = mvss / np.sqrt(3)

#define variables to graph
xs = Hz**(-0.5)
ys = mvsm / Hz
yes = mvse / Hz

# Fit model
def model(x, a, b):
    return a + b*x

#fitting weighted least squares line
popt, pcov = curve_fit(model, xs, ys, sigma=yes, absolute_sigma=True)
as_, bs = popt
sas, sbs = np.sqrt(np.diag(pcov))
#define line specific to each metal
xs_line = np.linspace(xs.min(), xs.max(), 200)
ys_line = model(xs_line, as_, bs)

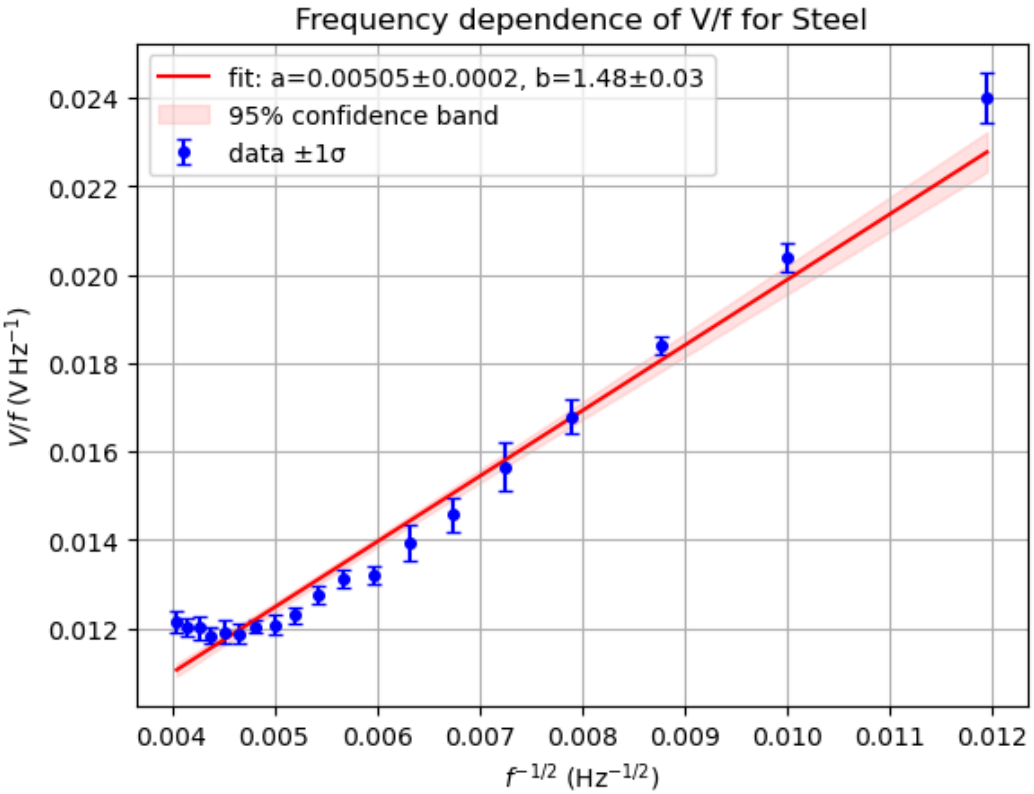
# Compute ±1 sigma fit uncertainty band
sas2, sbs2 = pcov[0, 0], pcov[1, 1]
sasb = pcov[0, 1]
err_s = np.sqrt(sas2 + (xs_line**2)*sbs2 + 2*xs_line*sasb)
```

```
# Compute 95% confidence band by multiplying standard error by 1.96
CIs = 1.96 * err_s

plt.errorbar(xs, ys, yerr=yes, fmt='o', capsize=3, ms=4,
             color='blue', label='data ±1σ')
plt.plot(xs_line, ys_line, 'r-',
         label=f'fit: a={as_:.3g}±{sas:.1g}, b={bs:.3g}±{sbs:.1g}')

#Plot 95 percent confidence interval
plt.fill_between(xs_line, ys_line - CIs, ys_line + CIs,
                 color='red', alpha=0.1, label='95% confidence band')

#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for Steel')
plt.legend()
plt.grid(True)
plt.show()
```



In [207…

```
#Plot all metals together
plt.figure(figsize=(7,5))

# Copper with error bars coloured and uncertaities
plt.errorbar(xc, yc, yerr=yec, fmt='o', capsize=3, ms=4,
             mfc='white', mec='blue', mew=1, label='Copper data ±1σ')
plt.plot(xc_line, yc_line, 'b-', label='Copper fit')
plt.fill_between(xc_line, yc_line - CIc, yc_line + CIc,
                 color='blue', alpha=0.1, label='Copper 95% CI')

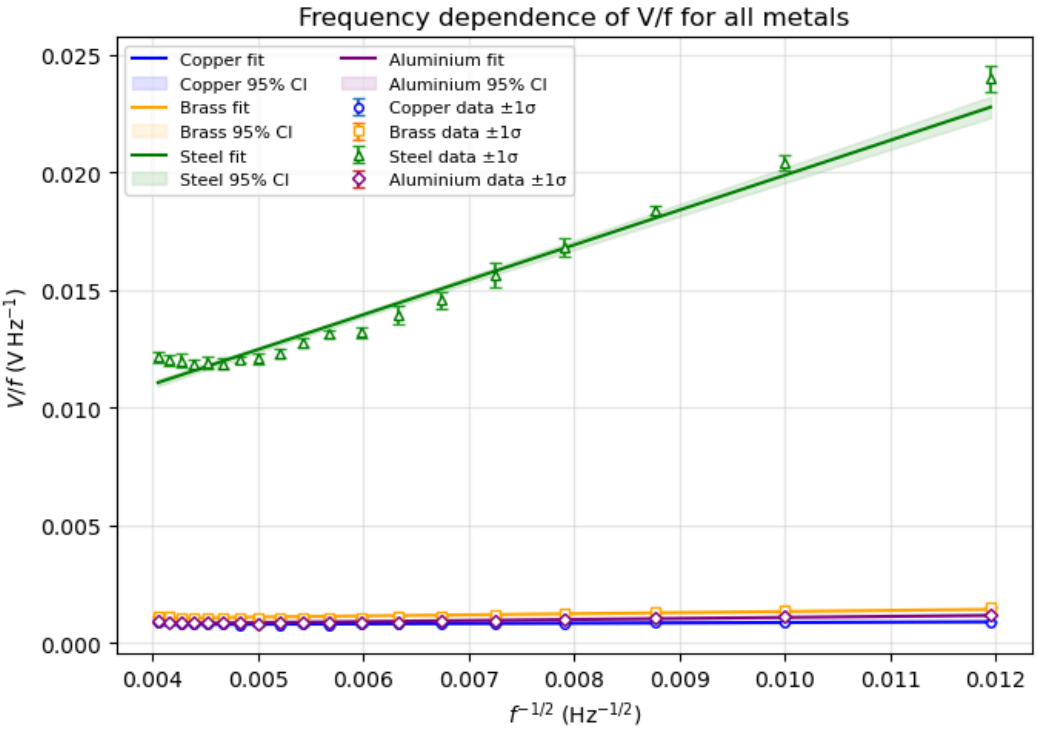
# Brass with error bars coloured and uncertaities
plt.errorbar(xb, yb, yerr=yeb, fmt='s', capsize=3, ms=4,
```

```
            mfc='white', mec='orange', mew=1, label='Brass data ±1σ')
plt.plot(xb_line, yb_line, 'orange', label='Brass fit')
plt.fill_between(xb_line, yb_line - CIb, yb_line + CIb,
                 color='orange', alpha=0.1, label='Brass 95% CI')

# Steel with error bars coloured and uncertaities
plt.errorbar(xs, ys, yerr=yes, fmt='^', capsize=3, ms=4,
             mfc='white', mec='green', mew=1, label='Steel data ±1σ')
plt.plot(xs_line, ys_line, 'g-', label='Steel fit')
plt.fill_between(xs_line, ys_line - CIs, ys_line + CIs,
                 color='green', alpha=0.1, label='Steel 95% CI')

# Aluminium  with error bars coloured and uncertaities
plt.errorbar(xa, ya, yerr=yea, fmt='D', capsize=3, ms=4,
             mfc='white', mec='purple', mew=1, label='Aluminium data ±1σ')
plt.plot(xa_line, ya_line, 'purple', label='Aluminium fit')
plt.fill_between(xa_line, ya_line - CIa, ya_line + CIa,
                 color='purple', alpha=0.1, label='Aluminium 95% CI')

#Plot graph with title and labels
plt.xlabel(r'$f^{-1/2}\ \mathrm{(Hz^{-1/2})}$')
plt.ylabel(r'$V/f\ \mathrm{(V\,Hz^{-1})}$')
plt.title('Frequency dependence of V/f for all metals')
plt.legend(fontsize=8, ncol=2)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



In [ ]:

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